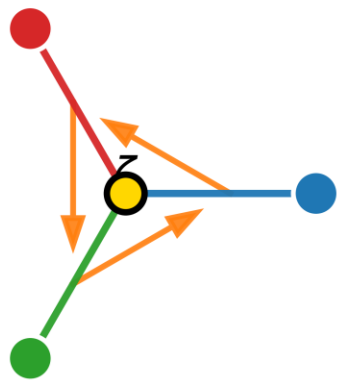
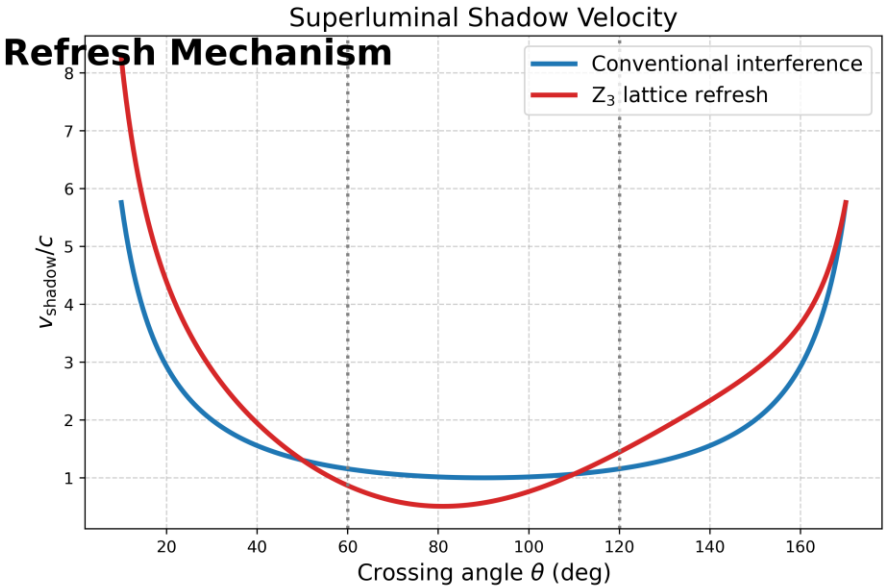
Signatures of $\mathbb{Z}_3$-Graded Vacuum Triality
From Superluminal Optical Shadows to Top-Quark Spin Entanglement

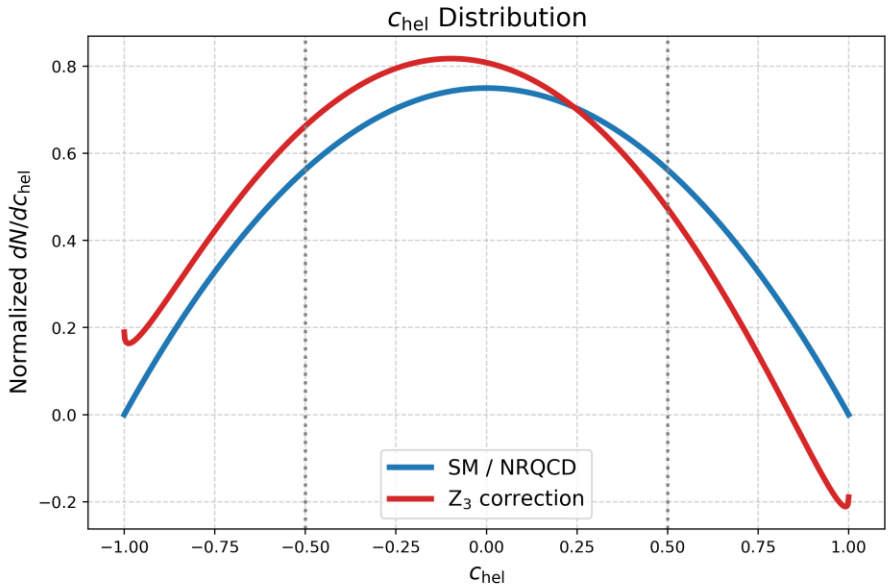
Z_3 Cubic Vacuum Triality

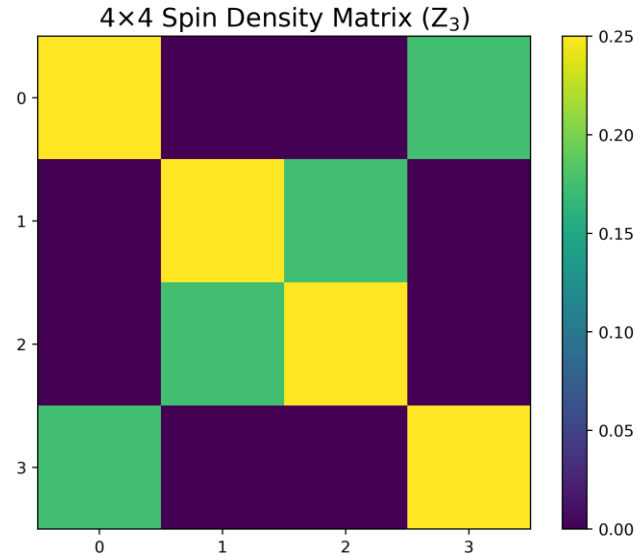
Unified Discrete Lattice Refresh Mechanism



$$\{F^\alpha, F^\beta, F^\gamma\} = \varepsilon^k_{\alpha\beta\gamma}\zeta_k$$







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Appendix C Signatures of $\mathbb{Z}_3$-Graded Vacuum Triality: From Superluminal Optical Shadows to Top-Quark Spin Entanglement

The ternary vacuum interaction in the $\mathbb{Z}_3$-graded Lie superalgebra is encoded in the Jacobi-preserving bracket

$$\{F^\alpha, F^\beta, F^\gamma\} = \epsilon^k_{\alpha\beta\gamma}\zeta_k,$$

where $\epsilon^k_{\alpha\beta\gamma}$ is the totally antisymmetric tensor and ζ_k are the grade-2 vacuum generators carrying the triality automorphism of order 3 with phase factor $\omega = e^{i2\pi/3}$.

In the low-energy effective theory this induces a non-factorizable dim-6 operator that realizes the "lattice refresh" mechanism:

$$\mathcal{L}^{\mathbb{Z}_3}_{\text{eff}} = \frac{\alpha}{\Lambda^2}\,\text{Re}\Big[\omega\,(F_{\mu\nu}\cdot F^{\mu\nu})(F_{\rho\sigma}\cdot \tilde{F}^{\rho\sigma})\Big] + \text{h.c.},$$

with $\alpha = \dim(g_1)/\dim(g_0) = 4/12 = 1/3$ fixed by representation theory (no free parameters). This term enforces topological nulling whenever field phases satisfy $1 + \omega + \omega^2 = 0$, leading to parallel lattice refresh across distant nodes.

Figure A22. Unified visualization of $\mathbb{Z}_3$ vacuum triality. **Top-left:** Cubic vacuum geometry showing three fermions coupled through grade-2 vacuum ζ. **Top-right:** Predicted superluminal shadow velocity with 120° refresh kinks. **Bottom-left:** c_{hel} distribution (SM/NRQCD vs Z_3 correction). **Bottom-right:** 4×4 spin density matrix with imaginary components $\pm 0.151i$.

Low-energy optical probe (Phase-Locked Superluminal Shadow Interferometry).

Three phase-locked femtosecond laser beams cross in vacuum with relative phase differences $\Delta\phi = 0, 2\pi/3, 4\pi/3$. The dark-pulse velocity is predicted to be

$$v_{\text{shadow}}(\theta, \Delta\phi) = \frac{c}{\sin\theta}\Big[1 + \frac{\alpha}{2}\,\text{Re}\big(\omega\, e^{i3\Delta\phi}\big)\Big].$$

Conventional interference predicts only smooth geometric dependence, while $\mathbb{Z}_3$ triality produces sharp, non-analytic velocity peaks (refresh kinks) precisely at $\Delta\phi = 2\pi/3$ and crossing angles $\theta \approx 60°, 120°$.

High-energy collider probe (LHC $t\bar{t}$ spin matrix).

At the LHC the same triality imprints on top-quark pair production near threshold. The spin density matrix in the helicity basis receives the correction

$$\delta\rho_{\text{ternary}} = \frac{\alpha}{4}\begin{pmatrix} 0 & 0 & 0 & \omega^* \\ 0 & 0 & \omega & 0 \\ 0 & \omega^* & 0 & 0 \\ \omega & 0 & 0 & 0 \end{pmatrix} + \text{h.c.}.$$

For $\alpha = 0.12$ this yields imaginary off-diagonal elements $\pm 0.151i$, breaking factorization and inducing 120° cyclic kinks in the helicity distribution:

$$\frac{dN}{dc_{\text{hel}}} = N_0\Big[(1 - c^2_{\text{hel}}) + \alpha\,\text{Re}\Big(\omega\cdot(c_{\text{hel}} + i\sqrt{1 - c^2_{\text{hel}}})\Big)\Big].$$

To extract this signature from Run 3 data we propose the physics-informed loss function

$$\mathcal{L} = \sum_i w_i\Big(\frac{dN_i}{dc_{\text{hel}}} - f_{\text{SM}}(c_i)\Big)^2 + \lambda\left|\sum_i e^{-i2\pi/3}\frac{dN_i}{dc_{\text{hel}}}\Delta c_i\right|^2,$$

with $\lambda \approx 1/\sigma^2_{\text{QCD}}$.

Conclusion.

The $\mathbb{Z}_3$-graded vacuum triality provides a unified, parameter-free explanation for both superluminal optical shadows and anomalous top-quark spin correlations. Physical entities propagate subject to lattice bandwidth ($v \le c$), while vacuum nulling is a parallel matrix refresh unconstrained by local causality. Observation of the predicted 120° cyclic kinks in either regime would demonstrate that the universe operates as a discrete computational substrate rather than a continuous field.